BRANDPILOT

Round 3 — Final Project Report

Intelligent Brand Governance & Content Review Platform

Project	BrandPilot
Competition	Codex 48 — Round 3
Developer	Adit Dinesh Shinde (Solo Project)
Technology	Python · Streamlit · OpenCV · SQLite
Repository	github.com/Adit24/BrandPilot
Live application	brandpilot-adit-codex48.streamlit.app

A practical prototype for reviewing brand-facing content, screening visual assets, and recording human review decisions.

1. Executive Summary

BrandPilot is a Streamlit-based brand governance prototype that helps teams review marketing copy against configurable brand guidance, inspect visual creatives for potential logo matches, and preserve a record of human review decisions. The Round 3 build extends the original text-review workflow with document ingestion, visual screening, and a SQLite-backed audit log.

The system is designed as a decision-support tool. Automated findings are indicators for human review, not final legal, brand, or compliance determinations.

2. Core Product Capabilities

Content and brand-guideline review

- Configurable brand workspace with brand name, guidelines, preferred and avoided terminology, and approved facts.
- Draft review using deterministic checks for terminology, risky absolute or promotional claims, unsupported numeric claims, hype language, and selected readability/accessibility heuristics.
- Findings include severity, quoted text, rationale, rule/source, suggested improvement, and confidence.
- Human decision workflow supports review status, reviewer comments, editable revisions, and re-review.

Document ingestion

- Accepts PDF, DOCX, TXT, and Markdown uploads.
- Extracted text is placed into an editable draft area before review.
- PDF extraction relies on selectable text; scanned-image OCR is not included. Long documents are capped by the configured input limit.

Visual brand screening

- Compares an uploaded approved reference logo with a marketing creative using OpenCV ORB feature detection and descriptor matching.
- Returns a match score, match counts, and a screening status to help flag creatives for manual inspection.
- A successful identical-logo test returned a 100% feature-match score with 1,493 good matches. This is a positive test only; it does not establish reliable detection across all logo variations or placements.

Review history and audit trail

- Review history is retained in the application’s local JSON history.
- SQLite audit records capture review identifier, reviewer label, selected role, action, decision, comment, and UTC timestamp.
- The reviewer identity and role are self-declared selections in this prototype; they are not authenticated accounts or enforced permissions.

3. Round 3 Improvements Delivered

Improvement	Implementation in this build
Visual Brand and Asset Verification	Implemented an initial logo-comparison workflow using OpenCV ORB feature matching. Results are screening signals and require human verification.
Secure Multi-User Review and Audit Trail	Implemented reviewer labels/role selection and SQLite audit event logging. Secure authentication, permission enforcement, and isolated workspaces remain future work.

Improvement	Implementation in this build
Document Ingestion and Content Workflow Integration	Implemented PDF/DOCX/TXT/MD text extraction into the review workflow. External webhook/CMS integration is not implemented in this build.

4. Technical Architecture

Layer	Technology / responsibility
User interface	Streamlit tabs, sidebar configuration, uploads, review controls, and history display
Review logic	Python rule-based checks and human decision workflow in app.py
Document service	PyPDF and python-docx extraction; UTF-8 text/Markdown reading
Visual service	Pillow image loading and OpenCV ORB feature matching
Audit storage	SQLite database with parameterized insert operations
Dependencies	Streamlit, google-genai (optional), Pillow, OpenCV headless, pypdf, python-docx

5. Testing and Verification

- Python compile check completed successfully for app.py, services, database, and modules.
- Application module import completed; Streamlit emitted expected bare-mode ScriptRunContext warnings because it was imported outside the Streamlit runner.
- Dependency consistency check completed with no broken requirements reported.
- PDF upload and selectable-text extraction were exercised with a sample multi-page marketing plan; extracted content appeared in the editable draft area.
- Visual comparison was tested with the same reference logo and creative, producing a 100% feature-match score.
- Review decision persistence and audit logging were tested; duplicate audit events caused by button-block indentation were corrected and re-tested.

These checks are functional spot tests, not a comprehensive automated test suite. A negative visual-match test and broader adversarial testing remain recommended.

6. Known Limitations and Responsible Use

- Rule-based content checks can produce false positives and false negatives; they do not understand every brand context.
- ORB feature matching is not a semantic logo-recognition model and can be affected by scaling, cropping, rotation, background, compression, and visual clutter.
- PDFs containing scanned images are not OCR-processed.
- Reviewer names and roles are self-declared. There is no secure login, role-based authorization, or tenant/workspace isolation.
- JSON and SQLite storage are local application storage. Persistence on a hosted ephemeral filesystem may not be guaranteed across restarts or redeployments.
- Gemini integration is optional and should only be considered active when a valid API key and successful API response are configured; the core deterministic review workflow does not depend on it.
- No external webhook or CMS integration is delivered in this version.

7. Future Roadmap

- Add authenticated users, enforced role-based permissions, and workspace-level data isolation.
- Move audit and review data to managed persistent storage with backup and migration support.
- Add OCR for scanned documents and richer layout-aware document extraction.
- Improve visual verification with logo-region detection, geometric verification, and broader positive/negative test datasets.
- Implement signed, authenticated webhooks and CMS integrations with structured review results.
- Add automated unit, integration, and regression tests, plus improved accessibility and usability validation.

8. Project Links

Live application: https://brandpilot-adit-codex48.streamlit.app/

Source repository: https://github.com/Adit24/BrandPilot

Round 3 source changes were committed and pushed to the main branch (commit d942b34).

Submission note: This report describes the implementation completed in the Round 3 prototype and distinguishes delivered functionality from planned work. Automated checks should be interpreted as assistance for human reviewers, not as guarantees of brand compliance.